

Causal Discovery in Cancer Gene Expression Data using Bayesian Networks

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INTRODUCTION

- Breast cancer begins when DNA mutations cause normal cells to become abnormal and divide uncontrollably independent of Estrogen.
- Patients are categorised into two classes:ER Positive and ER Negative. ER is an Estrogen Receptor protein found on both normal and cancer cells. In ER Positive patients,Estrogen connects to this protein and accelerates existing cancer cell division making the cancer grow aggressively.
- Hormone therapy works for ER Positive patients by blocking the ER protein and cutting off the Estrogen acceleration signal slowing cancer growth significantly.
- In ER Negative patients cancer cells produce little or no ER protein. Hormone therapy has nothing to block and completely fails leaving these patients with other treatment options.
- The Problem: Which gene is responsible for determining whether cancer cells produce the ER protein or not? This project uses Bayesian Networks to identify that causal gene.

DATASET AND OUTCOME VARIABLE

- We have used The Cancer Genome Atlas (TCGA) dataset, A publicly available real world database containing gene expression measurements from real breast cancer patients. Our dataset contains 705 patients with 1941 measurements per patient.
- After removing patients with unclear ER Status labels (Indeterminate, Not Performed) 549 patients remained for analysis in which 414 were ER Positive and 135 ER Negative. Only patients with confirmed ER Status were kept to ensure data reliability.
- Data type used: RNA-seq gene expression measuring how actively each gene is working in each patient. 604 RNA genes are available before filtering.
- Outcome variable: ER Status converted to binary ER Positive = 1 (hormone therapy works) and ER Negative = 0 (hormone therapy fails). This directly represents treatment response for each patient.
- Class imbalance: 414 Positive vs 135 Negative reflects the natural real world distribution of ER Status.

FILTERING THE REQUIRED GENES

- Problem: 604 RNA genes is too many for Bayesian Network. We Need to filter out the biologically relevant genes for our problem. We use two filters to do this process.
- Filter 1:MSigDB Curated Gene List: We Cross referenced 604 genes against known breast cancer gene database. Kept the genes already proven to be breast cancer related. We got 371 genes filtered out.
- Filter 2 :Limma Differential Expression Test: For each of the 371 genes,we measured the average activity across all 549 patients separately for ER Positive and ER Negative groups. Tested whether the difference between group averages is statistically real (adjusted p-value < 0.05). 314 significantly different genes were found after this process.
- Final Selection: Ranked 314 genes by p-value from most to least significant. Selected top 80 genes showing the strongest difference between ER Positive and ER Negative patients.
- Complete pipeline: 604 → 371 (MSigDB) → 314 (Limma) → 80 final genes. Every gene in the final 80 is both biologically validated AND statistically significant.

EMPERICAL BAYES T-TEST

- Linear Model: For each of the 371 genes, we use the limma library given by R, which fits $\text{Expression} = \beta_0 + \beta_1 \times \text{ER_Status} + \varepsilon$. β_0 is baseline activity in ER Negative patients. β_1 captures the difference in ER Positive patients. A large β_1 means the gene behaves very differently between groups making it a strong causal candidate.
- Empirical Bayes Moderation: Unlike a standard t-test which tests each gene independently limma tests all 371 genes simultaneously. It uses a moderated t-statistic: $t = (\text{mean ER Positive} - \text{mean ER Negative}) / \text{moderated standard error}$ where the standard error is stabilized using information borrowed from all 371 genes making results more reliable with limited samples.
- Benjamini-Hochberg FDR Control: Testing 371 genes simultaneously means $371 \times 0.05 = 18$ genes appear significant purely by chance. BH correction applies: BH threshold = $(\text{Rank} / 371) \times 0.05$ to each ranked p-value ensuring no more than 5% of kept genes are false discoveries. Result: 314 genuinely significant genes.
- Top 80 Selection: From 314 significant genes the top 80 with smallest adjusted p-values were selected. These show the strongest reliable difference between ER Positive and ER Negative patients while keeping the Bayesian Network computationally feasible.
- Every gene in the final 80 passed three levels of validation: Known breast cancer gene from MSigDB, statistically significant from Limma, and strongest evidence from top 80 selection.

DISCRETIZATION

- For our Problem, The Bayesian Network requires categorical data not continuous numbers. Gene expression values like 4.567 or 12.823 cannot be directly used. They must be converted into meaningful categories first.
- Each gene expression value is converted into three equal categories using quantile based cutoffs: Bottom 33% of values = Low, Middle 33% = Medium, Top 33% = High. This is applied independently for each gene across all 549 patients.
- Why three categories not two: Two categories creates a boundary problem where very similar values like 5.0 and 5.1 end up in completely opposite categories Low and High. Three categories keeps similar values together more accurately representing the continuous nature of gene activity.
- Why not more than three: With 549 patients divided into 80 genes having more than three categories reduces the number of patients per category. Three categories gives approximately 183 patients per category which is sufficient.
- Final result: 549 patients × 81 columns where all 80 gene columns are converted to Low Medium High categories and ER Status is converted to 0 and 1. The Data is now ready for Bayesian Network learning.

HILL CLIMBING WITH BIC SCORE

- Hill Climbing is a type of Bayesian Network which starts with an empty network. It tries adding removing or reversing one arrow at a time keeping only changes that improve the BIC score. Stops when no improvement is possible.
- $BIC = \log P(\text{Data}|\text{Network}) - (k/2) \times \log(N)$. Two competing parts balance each other to find the best network.
- $\log P(\text{Data}|\text{Network})$: For each gene calculates $P(\text{gene value} | \text{parent gene values})$ directly from patient counts. Example: $P(\text{ESR1=High} | \text{FOXA1=High}) = \text{patients where both are High} / \text{total patients where FOXA1=High}$. Summed across all 549 patients. Higher value means the arrow relationship holds consistently across patients.
- Penalty $(k/2) \times \log(N)$: k is the number of probability parameters estimated. Each new arrow increases k adding a cost of 3.15 BIC units $(\log(549)/2)$. Arrow is only kept if improvement in $\log P$ exceeds this penalty. Prevents overfitting by ensuring every arrow genuinely explains gene relationships not random noise.

TABU SEARCH WITH BDE SCORE

- Same as Hill Climbing but maintains a Tabu List of recently made moves that are forbidden. Forces exploration of new directions even if temporarily worsening the score allowing escape from local maxima. Always returns the BEST solution found across all explorations not just the current position.
- $BDe = \prod_i \prod_j [\Gamma(\alpha_{ij}) / \Gamma(\alpha_{ij} + N_{ij}) \times \prod_k \Gamma(\alpha_{ijk} + N_{ijk}) / \Gamma(\alpha_{ijk})]$. Measures how well the network explains the data but unlike BIC adds Dirichlet prior counts α_{ijk} before observing any patient data.
- Prior counts $\alpha_{ijk} = N' / (q_i \times r_i)$ where N' is equivalent sample size q_i is number of parent combinations and r_i is number of categories. These prior counts prevent overconfident conclusions from small samples by pulling extreme probability estimates towards more reliable values.
- Key difference from BIC: BIC calculates probabilities purely from observed patient counts. With only 549 patients some counts may be very small giving unreliable probabilities. BDe adds prior counts to every calculation ensuring no probability is based on too few patients making results more reliable.
- Combined advantage: Tabu Search escapes local maxima that trap Hill Climbing while BDe handles small sample uncertainty better than BIC. Together they produce a more reliable and thorough network than Hill Climbing with BIC alone.

BOOTSTRAP VALIDATION

- Problem: A single Bayesian Network built on our specific 549 patients might not be reliable. Bootstrap validation tests whether our findings are consistent across different patient samples.
- Process: Randomly resample 549 patients with replacement meaning some patients appear twice and some not at all. Build a new Bayesian Network on this resampled data. Repeat this 500 times producing 500 slightly different networks each built on a slightly different patient sample.
- Strength metric: For each possible gene connection count how many of the 500 networks contained that connection. $\text{Strength} = \text{number of networks containing connection} / 500$. Strength above 0.50 means the connection appeared in majority of networks and is considered reliable.
- Direction metric: For each connection that appeared count what proportion of those networks had the arrow pointing in a specific direction. $\text{Direction} = \text{proportion of networks agreeing on arrow direction}$. Direction above 0.80 means we are confident about which gene is influencing which.
- Why 500 replicates: With fewer replicates like 100 the strength and direction estimates are unstable and change significantly between runs. At 500 replicates the estimates stabilize meaning adding more replicates barely changes the results. 500 is the standard in published bioinformatics research balancing reliability with computational feasibility.

CONSENSUS NETWORK

- After 500 bootstrap networks we have 500 slightly different gene maps. Instead of trusting any single map we build one final reliable consensus network by keeping only arrows that appeared in more than 50% of 500 networks and removing everything below this threshold.
- Why 50% threshold: An arrow appearing in less than 50% of networks means more networks rejected it than accepted it. This arrow is unreliable and likely appeared due to the specific patients sampled not because the relationship is genuinely real.
- Results: Hill Climbing consensus kept 86 arrows from original 131. Tabu Search consensus kept 82 arrows from original 114. Both removed unreliable arrows that did not survive across different patient samples.
- Key finding: Both consensus networks independently identified ESR1 as the ONLY reliable direct parent of ER Status. AGTR1 was found in single network analysis but only appeared in 34.2% of bootstrap networks falling below the 50% threshold and was removed as unreliable.
- Significance: The consensus network is the final validated gene map. Every arrow in it appeared consistently across 500 different patient samples making it far more trustworthy than any single network built on one specific set of 549 patients.

BOOTSTRAP STRENGTH AND DIRECTION RESULTS

- ESR1 → ER Status confirmed by Hill Climbing Bootstrap: Strength = 0.876 meaning ESR1 → ER Status appeared in 87.6% of 500 networks. Direction = 0.953 meaning 95.3% of those networks agreed the arrow points FROM ESR1 TOWARD ER Status. Both metrics confirm this is a highly reliable and directionally confident finding.
- ESR1 → ER Status confirmed by Tabu Search Bootstrap: Strength = 0.894 meaning it appeared in 89.4% of 500 networks. Direction = 0.928 meaning 92.8% agreed on direction. Both algorithms independently confirm ESR1 → ER Status with high confidence making this an extremely robust finding.
- AGTR1 → ER Status rejected: Strength = 0.342 meaning it only appeared in 34.2% of networks falling below the 50% threshold. Although direction = 0.833 shows consistent directionality when it does appear the connection itself is too unstable across different patient samples to be considered reliable.
- FOXA1 ↔ ESR1 uncertain direction: Strength = 0.952 confirms a very strong connection at 95.2%. However direction = 0.418 vs 0.582 shows a 58/42 split meaning networks disagreed on which gene influences which. Reported as undirected edge as direction cannot be determined from observational data alone.
- Conclusion: ESR1 is the only gene confirmed as a reliable direct causal parent of ER Status across both algorithms and both bootstrap analyses. All other candidate genes either failed the strength threshold or had uncertain direction.

VALIDATION: BAYES NETWORK VS LASSO LOGISTIC REGRESSION

- Validation approach: 5 fold cross validation used to fairly compare both Bayesian Network algorithms against LASSO Logistic Regression. Same 80 genes and same 549 patients used for all three models. 549 patients split into 5 equal groups of 110 patients. Each group tested once while remaining 4 groups used for training. Average accuracy reported across all 5 folds.
- What LASSO does: LASSO Logistic Regression takes all 80 genes as input and finds which combination of genes best predicts ER Status. It shrinks irrelevant gene coefficients to zero automatically selecting the most predictive genes. It achieves 92% accuracy but only tells us which genes correlate with ER Status not which ones cause it.

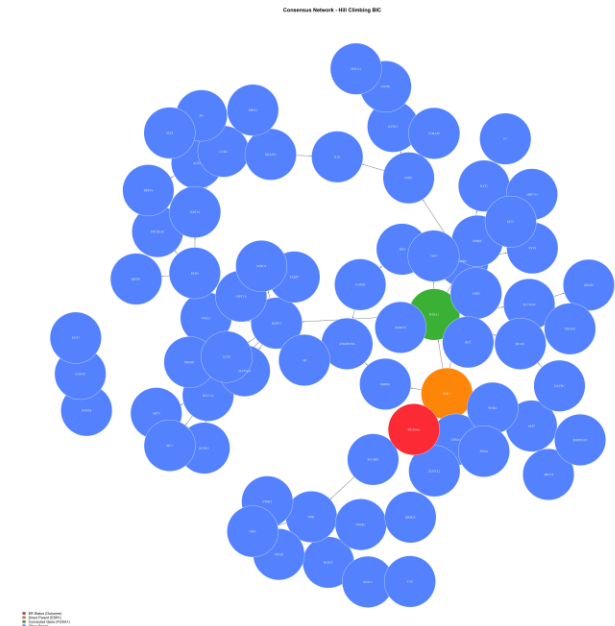
RESULTS

MODEL	FOLD-1	FOLD-2	FOLD-3	FOLD-4	FOLD-5	AVERAGE
HILL CLIMBING	91.8	86.4	87.3	90.0	87.2	88.5
TABU SEARCH	91.8	86.4	90.0	90.0	86.2	88.9
LOGISTIC LASSO	99.1	93.6	92.7	85.5	89.0	92.0

- All three models were evaluated using 5 fold cross validation on the same 80 genes and 549 patients ensuring a completely fair comparison.
- LASSO achieved the highest average accuracy of 92.0% by identifying correlations between all 80 genes and ER Status but provides no causal explanation.
- Hill Climbing BBN achieved 88.5% and Tabu Search BBN achieved 88.9% average accuracy. Both algorithms performed consistently across all 5 folds showing stable and reliable prediction.
- BBN performs only 3-4% below LASSO while additionally identifying ESR1 as the direct causal parent of ER Status with 87.6% to 89.4% bootstrap confidence across both algorithms. This causal finding is impossible to obtain from LASSO alone.
- The small accuracy trade off is justified. LASSO predicts who will be ER Negative. BBN predicts who will be ER Negative AND explains that ESR1 expression is the causal reason. This distinction is critical for identifying therapeutic drug targets in cancer research.

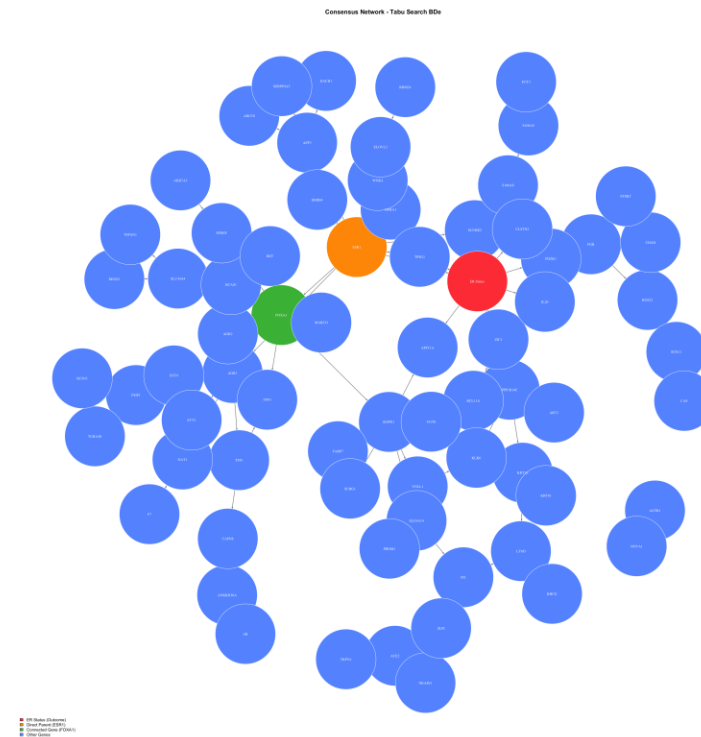
RESULTS-HILL CLIMBING BAYESIAN NETWORK

- The consensus network retained 86 reliable arrows from the original 131 after bootstrap validation. Only connections appearing in more than 50% of 500 bootstrap networks were kept ensuring every arrow shown is genuinely reliable.
- ER Status (red node) sits centrally in the network with ESR1 (orange node) as its only direct parent. This confirms ESR1 is the sole reliable direct causal gene determining ER Status with 87.6% bootstrap strength and 95.3% direction confidence.
- FOXA1 (green node) sits directly connected to ESR1 with 95.2% bootstrap strength. However direction was uncertain at 58/42 split meaning we cannot confirm from observational data alone whether FOXA1 influences ESR1 or ESR1 influences FOXA1.
- The majority of the 80 genes (blue nodes) form chains and clusters around the network periphery. They influence each other and indirectly connect to ER Status through intermediate genes but none of them directly point at ER Status with sufficient bootstrap confidence.
- The network structure confirms that ER Status is determined by a specific focused pathway rather than being influenced by all 80 genes simultaneously. This validates the causal discovery approach over standard correlation based methods.



RESULTS-TABU SEARCH BAYESIAN NETWORK

- The Tabu Search consensus network retained 82 reliable arrows from the original 114 after bootstrap validation. Slightly simpler than Hill Climbing confirming Tabu Search found a more parsimonious network structure by exploring a wider solution space.
- ER Status (red node) again has ESR1 (orange node) as its only direct parent. This independently confirms the Hill Climbing finding that ESR1 is the sole reliable direct causal gene determining ER Status with 89.4% bootstrap strength and 92.8% direction confidence.
- FOXA1 (green node) appears connected to ESR1 in the Tabu Search network as well. However Tabu Search found no reliable upstream parent for ESR1 in the consensus network confirming that what causes ESR1 to behave differently between patients cannot be determined from observational data alone.
- The overall network structure is slightly more spread out than Hill Climbing reflecting Tabu Search's ability to explore more network configurations. Despite exploring different paths both algorithms converged on the same key finding of $ESR1 \rightarrow ER\ Status$.
- The agreement between both consensus networks on ESR1 as the only reliable direct parent of ER Status is the strongest validation of our findings. Two completely different algorithms using different scoring functions and search strategies independently reached the same conclusion.



Conclusion

- Both Hill Climbing with BIC and Tabu Search with BDe consensus networks independently and consistently identified ESR1 as the only reliable direct causal parent of ER Status with 87.6% to 89.4% bootstrap strength and 92.8% to 95.3% direction confidence across 500 bootstrap replicates. This is the main finding of this project.
- BBN achieved 88-89% prediction accuracy compared to LASSO at 92% demonstrating that causal framing adds biological insight at only a 3-4% accuracy cost. LASSO predicts who will be ER Negative. BBN predicts who will be ER Negative AND identifies ESR1 as the causal reason providing a potential therapeutic target.
- Limitations: Findings are causal hypotheses not proven facts as observational data cannot definitively prove causation. 549 patients instead of proposed 1100 affects direction confidence. FOXA1 and ESR1 show strong connection at 95.2% but direction remains uncertain at 58/42 split. Expression data alone cannot capture ESR1 mutations which are the primary driver of treatment resistance requiring mutation data for complete analysis.
- Biological validation confirmed by two top journals. Bidard et al. 2025 New England Journal of Medicine confirmed ESR1 mutations directly linked to treatment resistance. Razavi et al. 2018 Cancer Cell confirmed ESR1 genomic alterations as primary drivers of endocrine resistance in breast cancer.
- Future work: Integrate ESR1 mutation data alongside expression data. Use larger patient cohort to improve direction confidence. Experimental validation of causal relationships identified. Investigate upstream regulators of ESR1 using longitudinal patient data to find what causes ESR1 to behave differently between patients.